

IAS PRELIMS/MAINS



SCIENCE & TECHNOLOGY



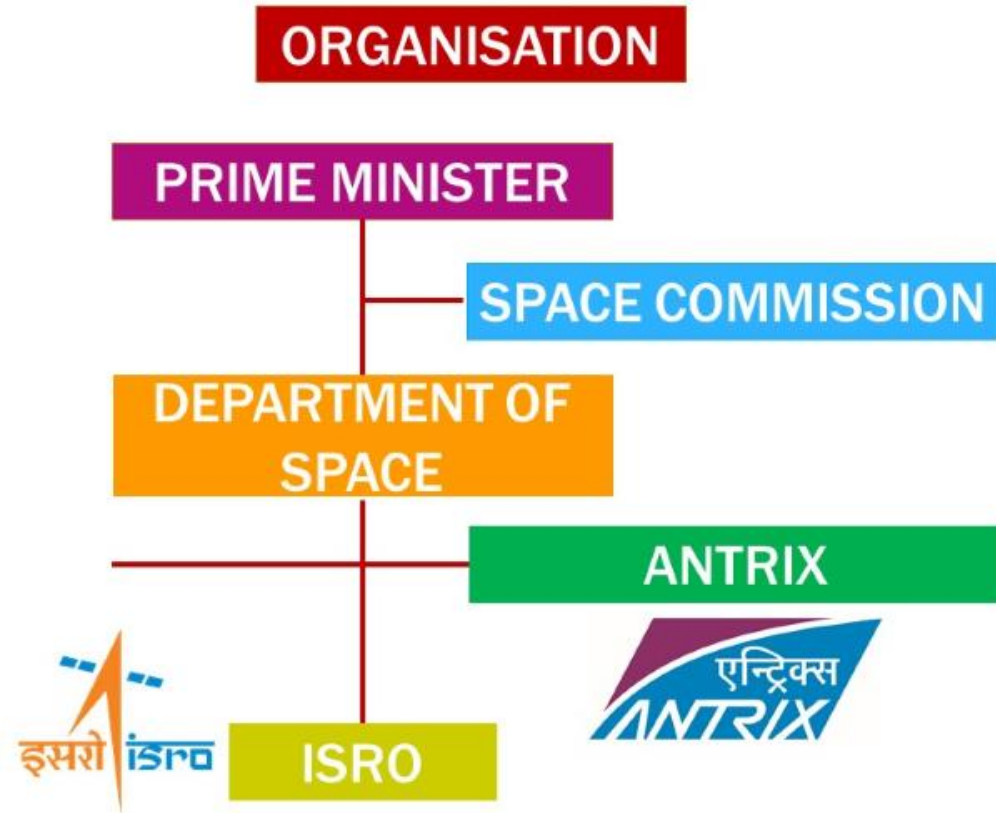
A COMPREHENSIVE COVERAGE

MODULE - 3

INDIAN SPACE RESEARCH ORGANISATION

INDIAN SPACE RESEARCH ORGANISATION

- The Indian Space Research Organization (ISRO) is the space agency of the Government of India headquartered in Bangalore.
- Its vision is to 'harness space technology for national development while pursuing space science research and planetary exploration'.
- It is managed by the Department of Space, which reports to the Prime Minister of India.



ANTRIX

- Antrix Corporation Limited (ACL), is a wholly owned Government of India Company under the administrative control of the Department of Space.
- Antrix Corporation Limited was established in 1992 as the commercial arm of ISRO for promotion of space products, technical consultancy services and transfer of technologies developed by ISRO.
- As the commercial and marketing arm of ISRO, Antrix is engaged in providing Space products and services to international customers worldwide.



SPACE COMMISSION

- It is the body that oversees the space program of India.
- Its members are senior civil servants like
 - ✓ Principal Secretary to PM
 - ✓ Cabinet Secretary
 - ✓ National Security Advisor
 - ✓ Foreign Secretary etc.
- It is headed by the Chairman of ISRO.
- It oversees the space program and provides advice to the PM.



IMPORTANT CENTRES OF ISRO

VIKRAM SARABHAI SPACE CENTRE (VSSC)

- VSSC is the main center of ISRO responsible for the design and development of launch vehicle technology.
- The Centre undertakes crucial responsibilities of design, manufacturing, analysis, development and testing of systems and technologies for launch vehicles.
- The major programmes at VSSC include Polar Satellite Launch Vehicle (PSLV), Geosynchronous Satellite Launch Vehicle (GSLV), Rohini Sounding Rockets, development of Geosynchronous Satellite Launch Vehicle (GSLV) Mk III, Reusable Launch Vehicles, air-breathing propulsion and critical technologies towards human spaceflight.



SOUNDING ROCKETS

- These rockets are solid propellant rockets used for studying the upper atmosphere and weather.
- It is an instrument carrying rocket designed to take measurements and perform scientific experiments in the upper layers of the atmosphere.
- They also serve as easy and affordable platforms to test new components and technologies intended for use in launch vehicles and satellites.

The launch of the first sounding rocket (imported from Russia) from Thumba near Thiruvananthapuram, Kerala in 1963, marked the beginning of the Indian Space Programme.

- In India all the sounding rocket activities are performed under the **Rohini Sounding Rocket Program**.
- ISRO started launching indigenously made sounding rockets from 1965 and experience gained was of immense value in the mastering of solid propellant technology.
- The sounding rocket programme was the bedrock on which launch vehicle technology could be built.



LIQUID PROPULSION SYSTEMS CENTRE (LPSC)

- It is the center for design and development of liquid propulsion stages for ISRO's Launch Vehicles. Key components of liquid propulsion systems are developed here.

SATISH DHAWAN SPACE CENTRE (SDSC), SRIHARIKOTA

- It is the Spaceport of India, is responsible for providing Launch Base Infrastructure for the Indian Space Programme.
- The Centre has two launch pads from where the rocket launching operations of PSLV and GSLV are carried out.
- The centre also produces solid propellant boosters for the launch vehicles of ISRO.



U R RAO SATELLITE CENTRE (URSC)

- Formerly known as ISRO Satellite Centre (ISAC) is the lead centre for building satellites and developing associated satellite technologies.
- These satellites are used for various applications like
 - ✓ Communication
 - ✓ Navigation
 - ✓ Meteorology
 - ✓ Remote Sensing
 - ✓ Space Science and
 - ✓ interplanetary explorations.



LAUNCH VEHICLES OF ISRO

PSLV- POLAR SATELLITE LAUNCH VEHICLE

- It is an expendable launch vehicle. It means it is used only once and its components are not recovered for re-use.
- The vehicle consists of several stages that are discarded one by one as the vehicle gains altitude.
- PSLV was developed by ISRO to allow it to launch remote sensing satellites into polar orbits. PSLV can also launch small satellites into Geostationary Orbits. PSLV was developed in the early 1990s at the Vikram Sarabhai Space Center at Trivandrum.
- PSLV was first launched in September 1993 but it was a failure. Then PSLV successfully completed its second mission in 1994. The 4th mission in 1997 was again a failure. But since then PSLV was launched more than 40 times with no failures. For Low Earth Orbits PSLV is widely used.
- Till now PSLV has launched more than 48 Indian and 210 foreign satellites, so it earned the title the 'Work horse of ISRO'.

PSLV- POLAR SATELLITE LAUNCH VEHICLE

- PSLV is a four stage rocket:
- **Fourth Stage: PS4**
 - ✓ The PS4 is the uppermost stage of PSLV, it uses a liquid fuel.
- **Third Stage: PS3**
 - ✓ The third stage of PSLV is a solid rocket motor that provides the upper stages high thrust after the atmospheric phase of the launch.
- **Second Stage: PS2**
 - ✓ PSLV uses a liquid rocket engine for its second stage, known as the Vikas engine, developed by Liquid Propulsion Systems Centre.
- **First Stage: PS1**
 - ✓ PSLV uses solid rocket motor in the 1st stage that is augmented by 6 solid strap-on boosters.



PSLV- POLAR SATELLITE LAUNCH VEHICLE

- ISRO has developed a number of variants of PSLV to cater to different mission requirements.
- There are currently two operational versions of the PSLV
 - ✓ the core-alone (PSLV-CA) without any strap-on motors, and
 - ✓ the PSLV-XL version, with six extended length (XL) strap-on motors.
- These configurations provide wide variations in payload capabilities up to 3,800 kg (8,400 lb) in LEO and 1,800 kg (4,000 lb) in polar orbit.
- **PSLV C37 (PSLV XL)** is the launch vehicle which was used to launch 104 satellites (101 foreign and 3 domestic) on 15th February 2017. The major satellite was the Cartosat-2D and the rest were nanosatellites.

CARTOSAT-2D:

- ✓ It is the 5th of the Cartosat series of satellites.
- ✓ These satellites are earth observation satellites built by India.
- ✓ Till now 6 Cartosat satellites have been launched by ISRO.
- ✓ The Cartosat series is part of the Indian Remote Sensing programme.

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